NIKUNJ GARG

Java Backend Developer | BCA Student

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SUMMARY

BCA student with hands-on experience developing backend applications using **Java 21**, **Spring Boot 3**, and **PostgreSQL**. Built projects involving **RESTful APIs**, database transactions, concurrency control, stateless **JWT** authentication, and real-time **WebSocket (STOMP)** communication. Interested in writing reliable, maintainable backend services with automated **JUnit 5** test coverage.

TECHNICAL SKILLS

- **Languages:** Java 21, SQL
- **Backend:** Spring Boot 3, Spring Data JPA, Hibernate ORM, RESTful APIs, WebSockets (STOMP)
- **Database:** PostgreSQL, Relational Schema Design (3NF), Composite B-Tree Indexing, ACID Transactions, Pessimistic Locking
- **Security & Testing:** Spring Security, JWT (Refresh Token Rotation), Role-Based Access Control (RBAC), JUnit 5, Mockito
- **Tools & DevOps:** Docker, Docker Compose, Git, GitHub, Maven, Postman, Render

PROJECTS

DevTinder — Developer Matching & Messaging Platform | Java 21, Spring Boot 3, PostgreSQL, WebSockets (STOMP), Spring Security, Docker GitHub Repo ↗

- Built bidirectional real-time messaging using **Spring WebSocket (STOMP)** paired with REST delta synchronization (`?after={id}`) and PostgreSQL persistence so users can retrieve missed messages upon reconnecting.
- Implemented a developer discovery and matching service that ranks candidate profiles dynamically by shared skill counts in descending order and tracks mutual interest using unique composite constraints on swipe records.
- Hardened authentication using stateless **Spring Security** filters, short-lived **JWT** access tokens, sliding-window IP rate limiting, and **Refresh Token Rotation (RTR)** with token family tracking to detect and revoke compromised sessions.
- Containerized the backend using a multi-stage **Dockerfile** and configured environment variables and health-check endpoints (`/api/health`) for cloud deployment integrating with a Next.js frontend.

LibroSphere — Library Management & Reservation System | Java 21, Spring Boot 3, PostgreSQL, Spring Data JPA, Spring Security, Docker, JUnit 5 GitHub Repo ↗

- Implemented pessimistic row-level locking (`SELECT ... FOR UPDATE`) in **Spring Data JPA** within atomic `@Transactional` boundaries to prevent race conditions and inventory overselling during concurrent checkouts.
- Verified thread safety and ACID resilience with an automated **JUnit 5** integration test using `CountDownLatch` and `ExecutorService`, proving 10 concurrent threads competing for 1 copy produced exactly 1 loan and 9 rejections with zero underflow.
- Architected an automated FIFO reservation waitlist that intercepts returned books, holding them exclusively for queued patrons with a 48-hour pickup window before restoring unclaimed inventory to general circulation.
- Built an automated nightly cron worker (`@Scheduled`) calculating tiered overdue penalties and added composite B-Tree indexes on `(returned, due_date)` and `(book_id, status, reserved_at)` to avoid full table scans.

AgriSathi AI — Agricultural Assistance Platform | Java 21, Spring Boot 3, PostgreSQL, Spring Security, Swagger 3.0, Cloudinary GitHub Repo ↗

- Developed 20+ **RESTful endpoints** in Spring Boot with Jakarta Validation and centralized `@ControllerAdvice` error handling for crop tracking, agronomic advisory, and marketplace operations.
- Built a multipart image upload pipeline with file size (<5MB) and MIME-type validation, streaming crop photos to Cloudinary CDN and returning automated disease diagnoses, confidence scores, and treatment plans.
- Configured stateless **JWT authentication** with **Role-Based Access Control (RBAC)** to separate farmer and merchant workflows, documenting all API contracts with interactive **OpenAPI / Swagger 3.0**.

EDUCATION

Bachelor of Computer Applications (BCA) — Shri Guru Ram Rai University, Dehradun **2025 – 2028**
Cumulative CGPA: 7.48 / 10.0 • **Semester 2 SGPA:** 7.96 / 10.0

LEADERSHIP & ACHIEVEMENTS

- **Secretary — TechMantra Coding Club:** Promoted from Core Member to Secretary within 10 months; organized 5+ hands-on coding bootcamps and technical workshops for 120+ student developers.
- **Startup Innovation Recognition:** Awarded Certificate of Recognition from **UCOST & SGRR University IIC** for pitching the startup concept *College Clash* in university innovation competitions.
- **Smart India Hackathon (SIH'26 Internal Round):** Awarded Certificate of Participation, collaborating in a collegiate software team to architect solutions for national-level problem statements.
- **Hackathon Competitions:** Awarded Certificate of Participation in **Algothon'26 (Algoxilla)**; engineered and deployed DevTinder backend in 72 hours during **Devlynix Buildathon 2.0**.
- **Algorithmic Problem Solving:** Solved data structures and algorithms problems in Java on LeetCode (@Nikunjgarg12), focusing on arrays, strings, two-pointers, and linked lists.